BuildMind

Project Requirements Document

AI-Based Residential House Design and Construction Planning System

1. Project Overview

BuildMind is an AI-based residential house design and construction planning system. The user provides land details, house requirements and budget. The system uses these inputs to generate simple house design concepts, 2D floor plans, a basic 3D view and an estimated budget. An AI chatbot will also help collect requirements and allow the user to request simple design changes.

2. Project Scope

- Collect user land and house requirements.
- Use an AI chatbot to understand and confirm user needs.
- Generate simple automatic 2D floor-plan concepts.
- Provide more than one design option where possible.
- Show a basic 3D visualization of the selected design.
- Provide a basic construction cost estimate based on floor area and budget.
- Allow simple user changes such as increasing a room size or adding/removing a feature.

3. User Input Requirements

Input Field	Description
Land Width	Width of the land, for example 40 ft
Land Length	Length of the land, for example 60 ft
Measurement Unit	Feet / Metres / Perches
Road Side / Direction	North / South / East / West where available
Number of Floors	1 / 2 / 3
Bedrooms	Required number of bedrooms
Bathrooms	Required number of bathrooms
Living Room	Yes / No
Dining Room	Yes / No
Kitchen Type	Open / Closed / Standard
Parking	Number of vehicles
Garden	Yes / No
Balcony	Yes / No
Other Rooms	Office, store room, prayer room, utility room, etc.

Preferred Style	Modern / Minimal / Traditional / Other
Budget	Estimated user budget in LKR

4. Expected System Outputs

Output	Description
Requirement Summary	A clear summary of the user inputs collected by the form or AI chatbot.
2D Floor Plan	A simple room layout showing room positions and approximate dimensions.
Design Options	Two or more layout options such as Budget, Balanced and Spacious where possible.
3D View	A basic 3D visualization created from the selected 2D layout.
Budget Estimate	Approximate cost and an indication of whether the design is within the user budget.
AI Suggestions	Simple suggestions to improve space usage or reduce cost.
Design Modification	Simple user-requested changes handled through the AI chatbot and planning logic.

5. Basic System Workflow

User Details → AI Requirement Understanding → Planning Rules → 2D Plan Generation → Design Selection → 3D Visualization → Budget Estimate → AI-Based Modification

6. 7-Member Team Responsibilities

Member	Main Role	Responsibilities
Member 1	Team Lead & System Integration	Project coordination, GitHub, task tracking, final integration and demo preparation.
Member 2	Construction Domain Research & Documentation	Research basic house-planning rules, collect expert feedback, document requirements and limitations.
Member 3	Frontend Developer	Build the Angular user interface, forms, dashboard, chatbot screen and design result pages.
Member 4	Backend & Database Developer	Create APIs, project data storage, authentication and communication between frontend and other modules.

Member 5	AI Chatbot Developer	Create requirement collection, prompt logic, structured data extraction and simple modification requests.
Member 6	2D Floor-Plan / Algorithm Developer	Create room placement logic, dimensions, boundary checks, overlap checks and multiple layout generation.
Member 7	3D Visualization Developer	Convert 2D layout information into a basic 3D view using Three.js or a similar library.

7. Suggested Technology Stack

Frontend: Angular

Backend: Python FastAPI

Database: SQLite for early development; PostgreSQL if needed later

AI: LLM API for chatbot and requirement understanding

2D Plan: Custom planning algorithm with coordinate-based room placement

3D: Three.js

8. Current Project Limitations

- The generated floor plan is a concept design and not a construction-ready engineering drawing.
- Structural calculations such as foundation, beam, column and reinforcement design are outside the first version.
- Official building approval and full regulation verification are outside the first version.
- Budget values are approximate and must be verified by a construction professional or quantity surveyor.
- Final construction decisions should be reviewed by qualified architects and engineers.

9. Minimum Success Criteria for the 1–1.5 Month MVP

1. User can enter land size, rooms, facilities and budget.
2. AI chatbot can understand and summarize the requirements.
3. System can generate at least one valid simple 2D floor plan.
4. System can display a basic 3D version of the selected plan.
5. System can provide a simple budget estimate.
6. The complete demo works from user input to final result without major errors.

10. Team Member Details

No.	Member Name	UT Number
1		
2		
3		
4		
5		
6		
7		